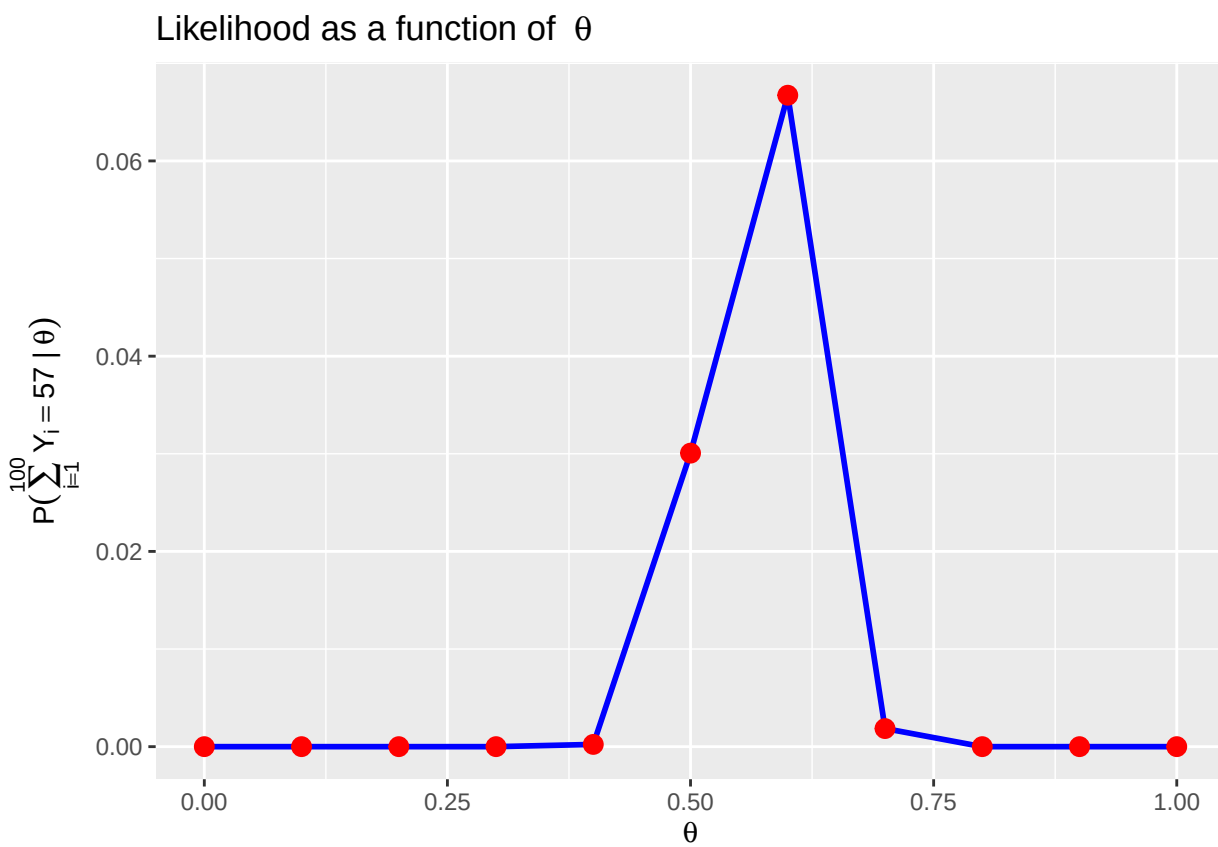
CHAP-3_BAYESIAN

Tawhid Tasdique

2025-12-19

Ex-3.1(b)

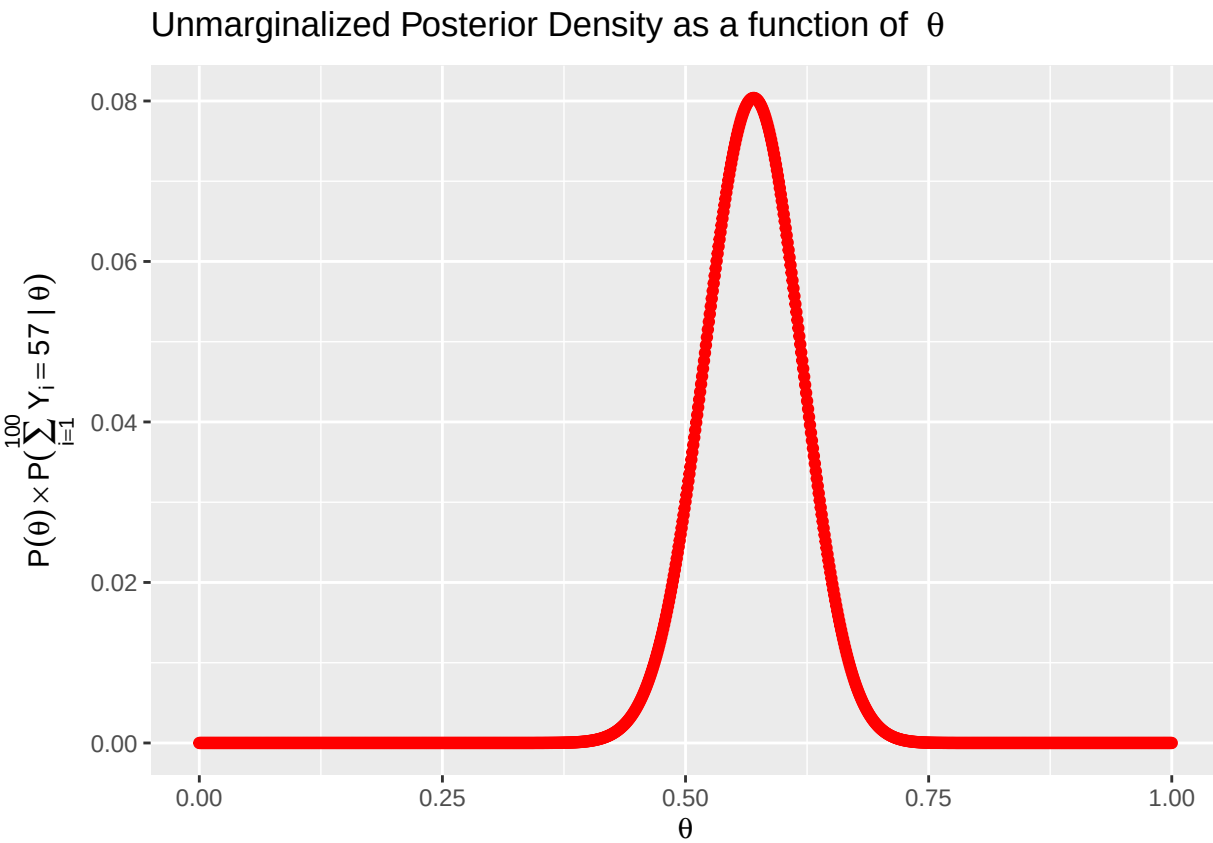
```r
# dataframe
dta <- tibble(theta = (0:10)/10) %>%
  mutate(
    likelihood = dbinom(
      x = 57,
      size = 100,
      prob = theta
      )
  )

# graphics
dta %>%
  ggplot(aes(x = theta, y = likelihood)) +
  geom_line(linewidth = 1, color = "blue") +
  geom_point(size = 3, color = "red") +
  xlab(expression(theta)) +
  ylab(expression(P(sum(Y[i], i == 1, 100) == 57 ~ "|" ~ theta))) +
  labs(
    title = expression("Likelihood as a function of " ~ theta)
  )
```



Ex-3.1(d)

When the prior belief follows the standard uniform distribution, the unmarginalized posterior distribution is equal to the likelihood of the data.

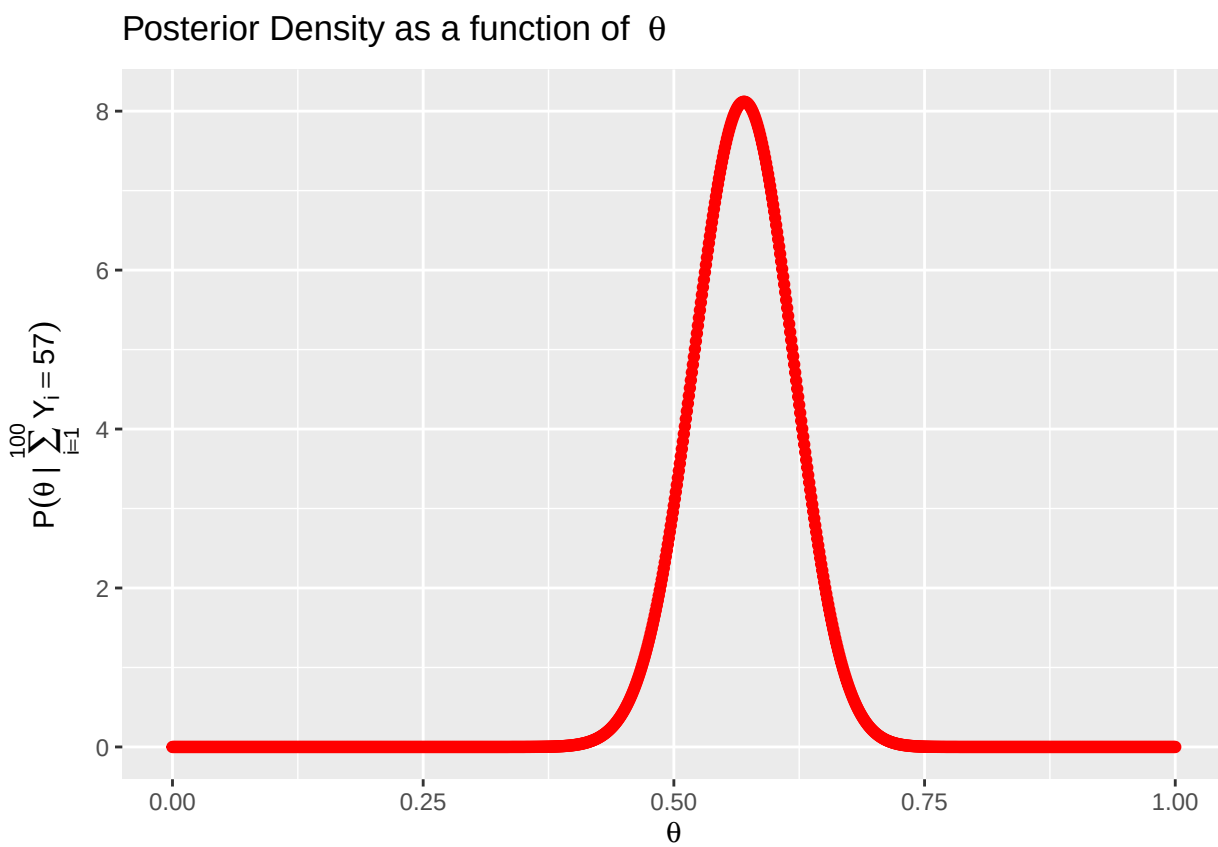
```
# dataframe
dta <- tibble(theta = (0:1000)/1000) %>%
  mutate(
    unmarginalized_posterior = dbinom(x = 57,
                                      size = 100,
                                      prob = theta)
  )

# graphics
dta %>%
  ggplot(aes(x = theta, y = unmarginalized_posterior)) +
  geom_point(size = 1.5, color = "red") +
  xlab(expression(theta)) +
  ylab(expression(P(theta) %*% P(sum(Y[i], i == 1, 100) == 57 ~ "|" ~ theta))) +
  labs(
    title = expression("Unmarginalized Posterior Density as a function of " ~ theta)
  )
```



Ex-3.1(e)

Plot (b) is a discrete version of the plot (d). And, the plot (e) is a scaled version of the plot (d). That's all I have to comment on the relationship among these graphs.

```
# dataframe
dta <- tibble(theta = (0:1000)/1000) %>%
  mutate(
    posterior = dbeta(x = theta,
                      shape1 = (1+57),
                      shape2 = (1+100-57))
  )

# graphics
dta %>%
  ggplot(aes(x = theta, y = posterior)) +
  geom_point(size = 1.5, color = "red") +
  xlab(expression(theta)) +
  ylab(expression(P(theta ~ "|" ~ sum(Y[i], i == 1, 100) == 57))) +
  labs(
    title = expression("Posterior Density as a function of " ~ theta)
  )
```



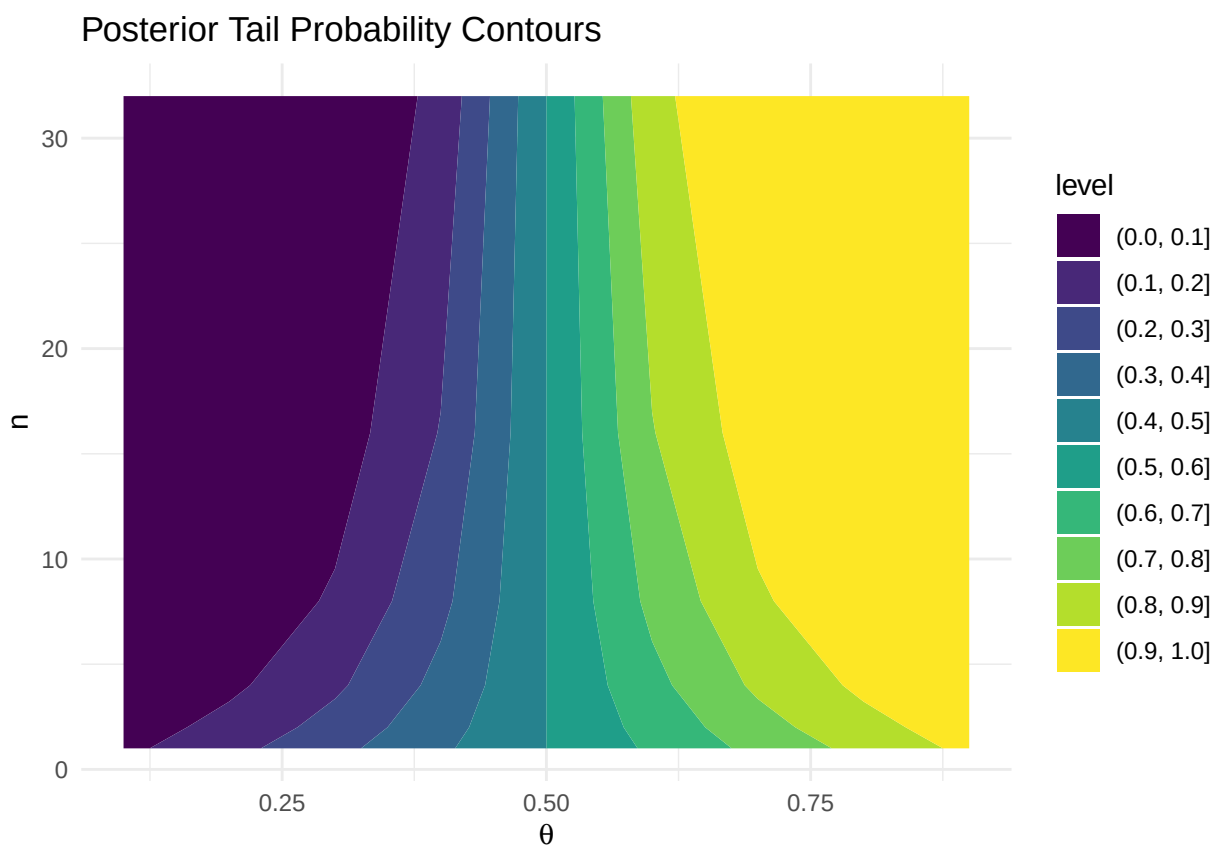
Ex-3.2

Problems I faced

If your x and y are deterministic functions of each other, contours cannot exist. The x and y axes must be independent for the contour plot to show. I can't give a better explanation at the moment. This is all I got from ChatGPT. So, `a` and `b` were not independent because of their formula, because of how they were constructed. That's why I kept getting an empty contour plot. The parameters θ and `n` on the other hand, were mutually independent. That's why the contour plot worked.

```
theta0 <- c(1:9)/10
n0 <- c(1, 2, 4, 8, 16, 32)

a <- outer(theta0, n0, `*`)
b <- outer(1 - theta0, n0, `*`)

dta <- expand.grid(
  theta = theta0,
  n = n0
) %>%
  mutate(
    shape = theta * n,
    rate  = (1 - theta) * n,
    prob  = pbeta(
```

```r
      q = 0.5,
      shape1 = shape,
      shape2 = rate,
      lower.tail = FALSE
    )
  )

dta %>%
  ggplot(aes(x = theta, y = n, z = prob)) +
  geom_contour_filled() +
  labs(
    x = expression(theta),
    y = "n",
    title = "Posterior Tail Probability Contours"
  ) +
  theme_minimal()
```

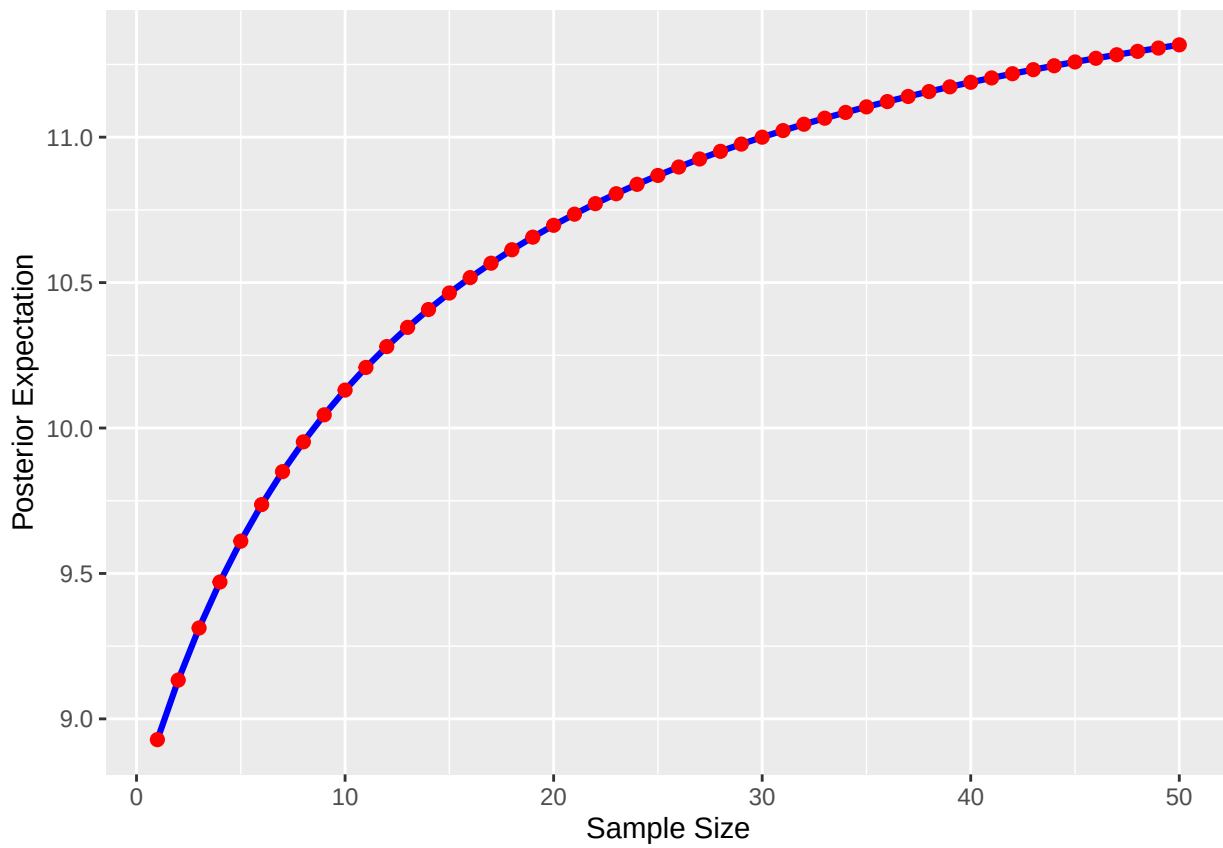


Ex-3.3(b)

```r
n0 <- 1:50

# never use lapply() if you can avail with vectorized operations
post_mean <- (12 * n0 + 113) / (n0 + 13)
```

```r
tibble(sample_size = n0, posterior_mean = post_mean) %>%
  ggplot(aes(x = sample_size, y = posterior_mean)) +
  geom_line(size = 1.1,
            color = "blue") +
  geom_point(color = "red",
             size = 2) +
  xlab("Sample Size") +
  ylab("Posterior Expectation")
```

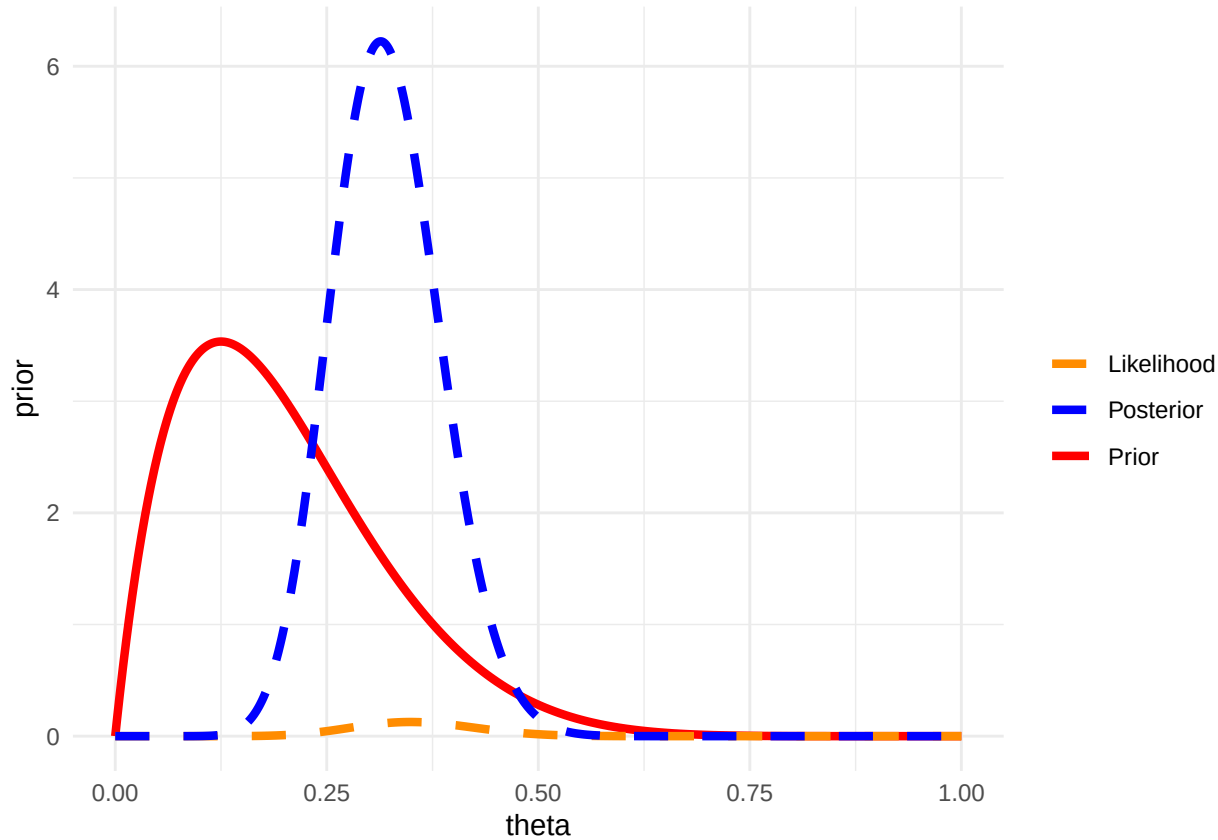
```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



Ex-3.4(a)

```r
cons <- choose(43, 15)

dta <- tibble(theta = seq(0,1000)/1000) %>%
  mutate(prior = dbeta(x = theta, shape1 = 2, shape2 = 8),
```

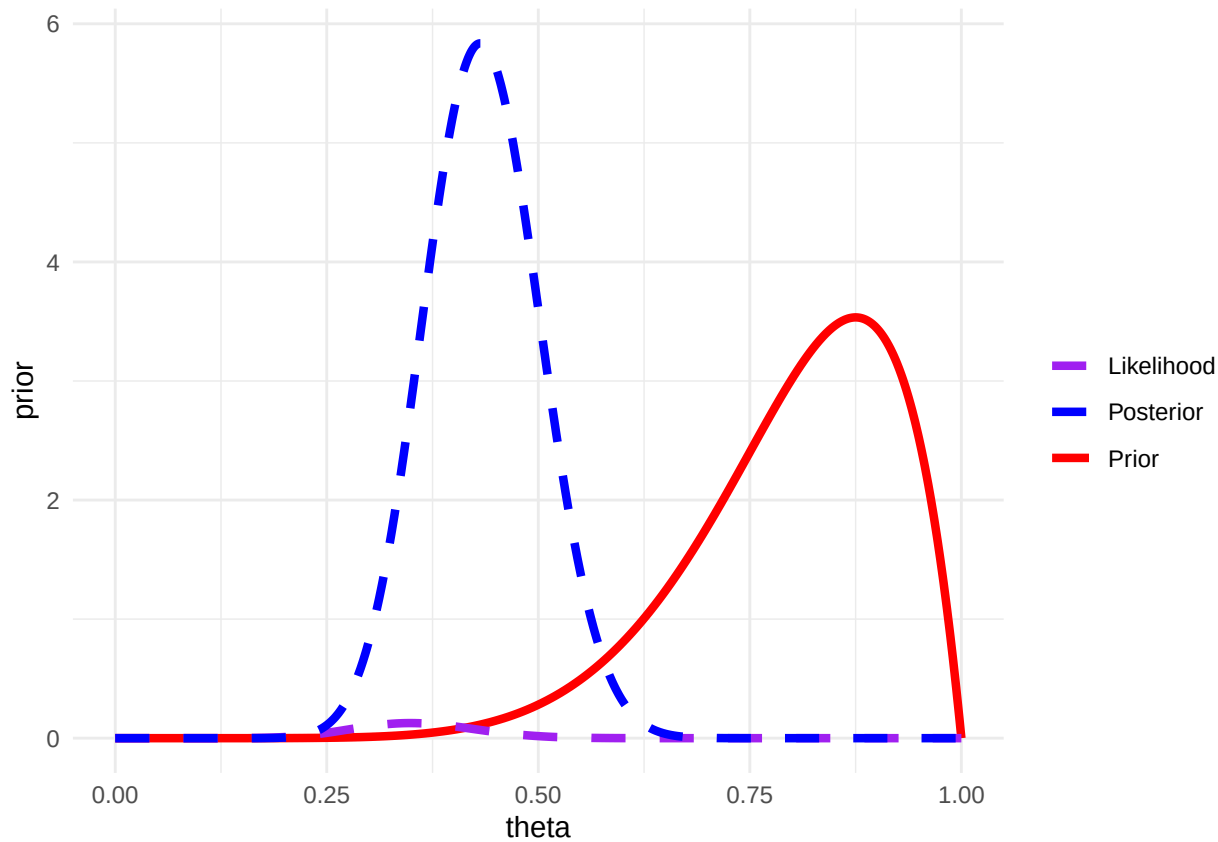
```r
         likelihood = cons*(theta^15)*((1-theta)^28),
         posterior = dbeta(x = theta, shape1 = 17, shape2 = 36))

dta %>%
  ggplot(aes(x = theta)) +
  geom_line(aes(y = prior, color = "Prior", linetype = "Prior"),
            size = 1.5) +
  geom_line(aes(y = likelihood, color = "Likelihood", linetype = "Likelihood"),
            size = 1.5) +
  geom_line(aes(y = posterior, color = "Posterior", linetype = "Posterior"),
            size = 1.5) +
  scale_color_manual(
    values = c(
      "Prior" = "red",
      "Likelihood" = "darkorange",
      "Posterior" = "blue"
    )
  ) +
  scale_linetype_manual(
    values = c(
      "Prior" = "solid",
      "Likelihood" = "dashed",
      "Posterior" = "dashed"
    )
  ) +
  labs(color = "", linetype = "") +
  theme_minimal()
```



Ex-3.4(b)

```
cons <- choose(43, 15)

dta <- tibble(theta = seq(0,1000)/1000) %>%
  mutate(prior = dbeta(x = theta, shape1 = 8, shape2 = 2),
         likelihood = cons*(theta^15)*((1-theta)^28),
         posterior = dbeta(x = theta, shape1 = 23, shape2 = 30))

dta %>%
  ggplot(aes(x = theta)) +
  geom_line(aes(y = prior, color = "Prior", linetype = "Prior"),
            size = 1.5) +
  geom_line(aes(y = likelihood, color = "Likelihood", linetype = "Likelihood"),
            size = 1.5) +
  geom_line(aes(y = posterior, color = "Posterior", linetype = "Posterior"),
            size = 1.5) +
  scale_color_manual(
    values = c(
      "Prior" = "red",
      "Likelihood" = "purple",
      "Posterior" = "blue"
    )
```

```
  ) +
  scale_linetype_manual(
    values = c(
      "Prior" = "solid",
      "Likelihood" = "dashed",
      "Posterior" = "dashed"
    )
  ) +
  labs(color = "", linetype = "") +
  theme_minimal()
```



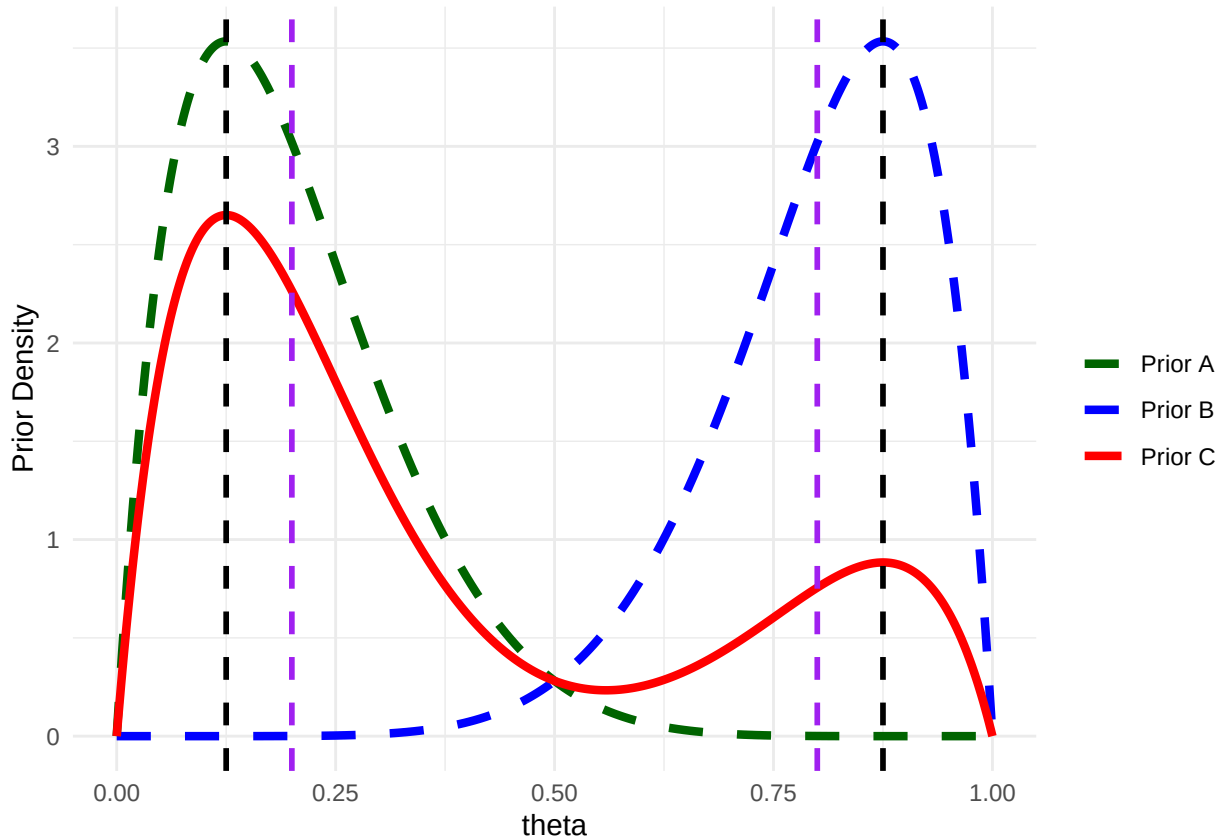
Ex-3.4(c)

Bimodal Prior: Interpretation

The following prior distribution has two peaks, indicating a belief that the recidivism rate is either high or low, rather than being centered around a moderate value like the priors in a) and b).

```
cons <- choose(43, 15) # to save time

# data frame
dta <- tibble(theta = seq(0,1000)/1000) %>%
  mutate(
```

```r
    priorC = 0.75*dbeta(x = theta, shape1 = 2, shape2 = 8) + 0.25*dbeta(x = theta, shape1 = 8, shape2 =
    priorB = dbeta(x = theta, shape1 = 8, shape2 = 2),
    priorA = dbeta(x = theta, shape1 = 2, shape2 = 8)
    )

# plots
dta %>%
  ggplot(aes(x = theta)) +
  geom_line(aes(y = priorA, color = "Prior A", linetype = "Prior A"),
            size = 1.5) +
  geom_line(aes(y = priorB, color = "Prior B", linetype = "Prior B"),
            size = 1.5) +
  geom_line(aes(y = priorC, color = "Prior C", linetype = "Prior C"),
            size = 1.5) +
  scale_color_manual(
    values = c(
      "Prior A" = "darkgreen",
      "Prior B" = "blue",
      "Prior C" = "red"
    )
  ) +
  scale_linetype_manual(
    values = c(
      "Prior A" = "dashed",
      "Prior B" = "dashed",
      "Prior C" = "solid"
    )
  ) +
  labs(y = "Prior Density", color = "", linetype = "") +
  theme_minimal() +
  geom_vline(xintercept = 0.8, # mean of prior B
             linetype = "dashed",
             linewidth = 1,
             color = "purple") +
  geom_vline(xintercept = 0.2, # mean of prior A
             linetype = "dashed",
             linewidth = 1,
             color = "purple") +
  geom_vline(xintercept = 0.875, # mode of prior B
             linetype = "dashed",
             linewidth = 1) +
  geom_vline(xintercept = 0.125, # mode of prior A
             linetype = "dashed",
             linewidth = 1)
```



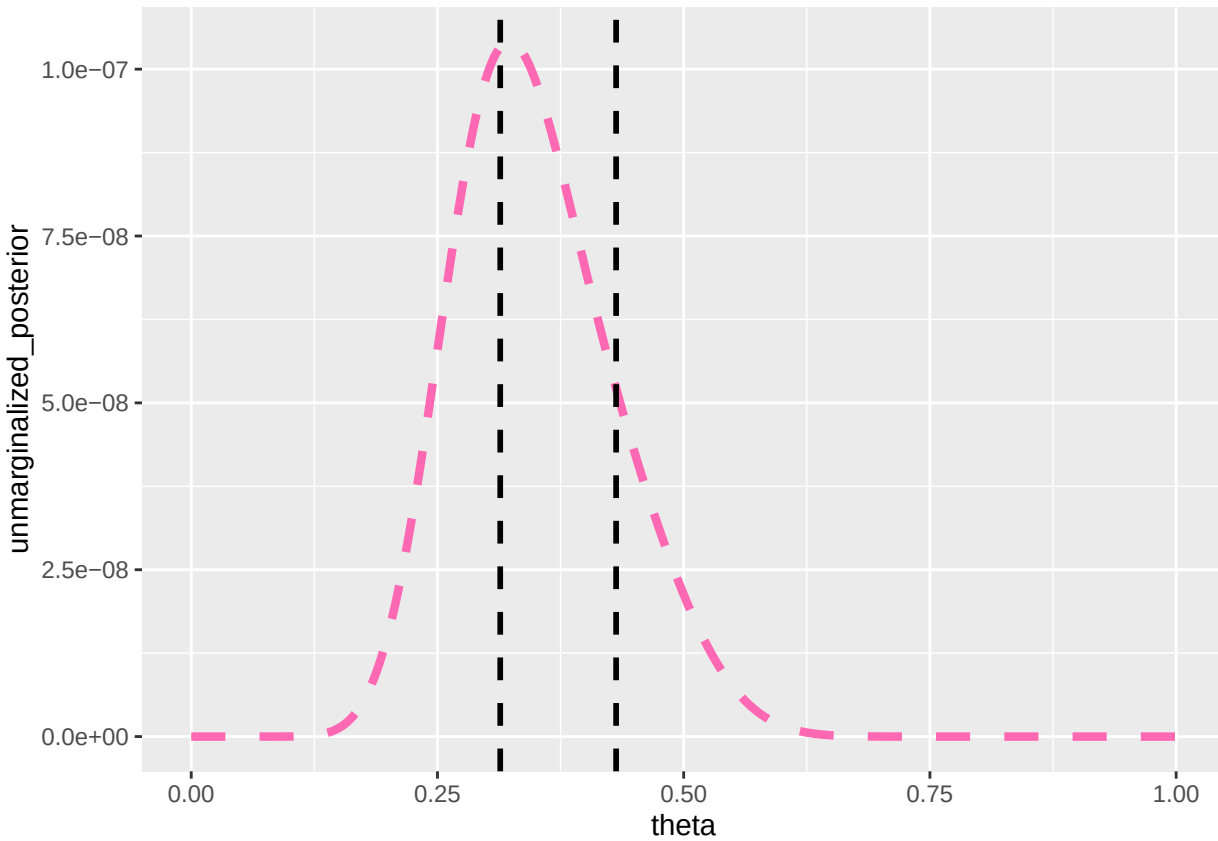
Ex-3.4(d)(iii)

Approximating the Mode of Mixed Posterior Distribution

The mode of the following mixed posterior distribution appears closer to that of the posterior A and farther from that of the posterior B.

```
# constant
cons <- choose(43, 15)*beta(17, 53)/4*beta(2, 8)

dta <- tibble(theta = seq(0, 1000)/1000) %>%
  mutate(
    unmarginalized_posterior = cons*(3*dbeta(x = theta, shape1 = 17, shape2 = 36) + dbeta(x = theta, sha


dta %>%
  ggplot(aes(x = theta, y = unmarginalized_posterior)) +
  geom_line(color = "hotpink",
            linewidth = 1.5,
            linetype = "dashed") +
  geom_vline(xintercept = (22/51), # mode of posterior B
             linetype = "dashed",
             linewidth = 1) +
  geom_vline(xintercept = (16/51), # mode of posterior A
```

```
              linetype = "dashed",
              linewidth = 1)
```

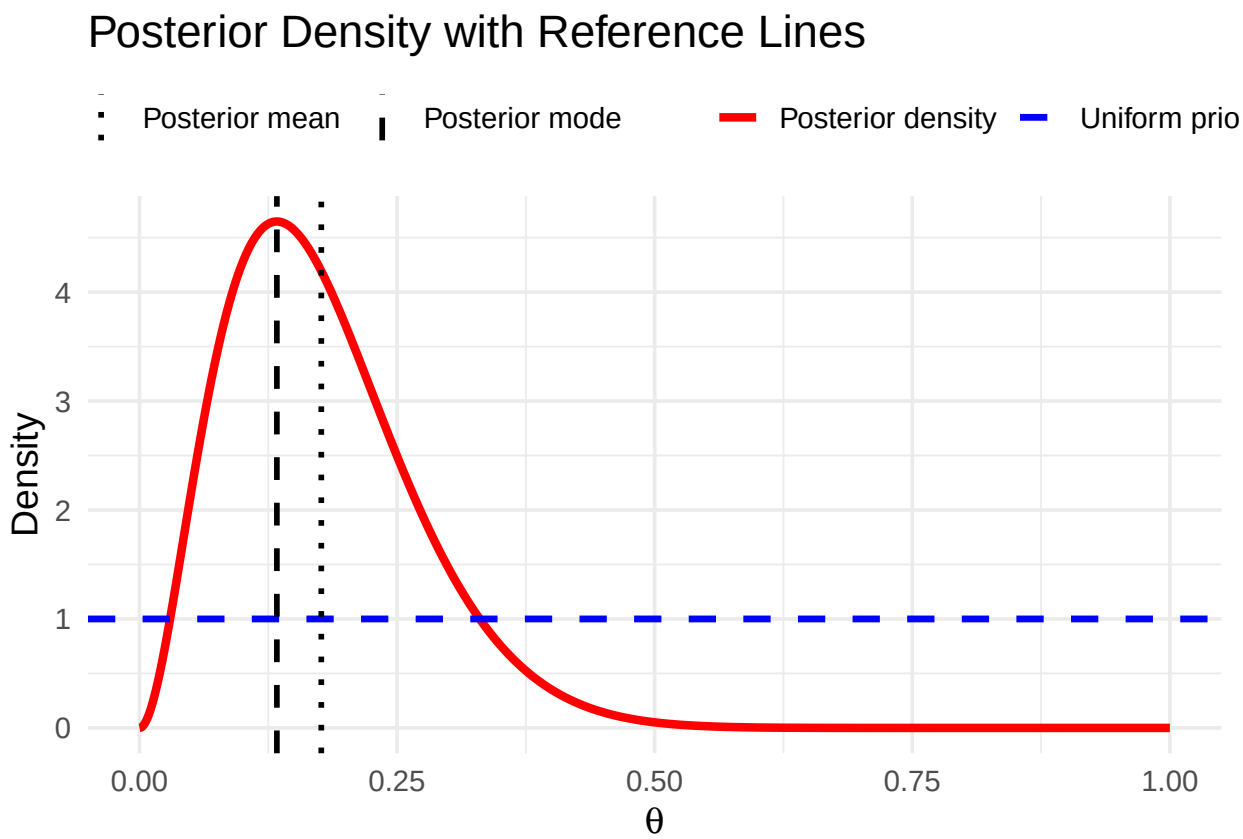



```
# How come it unimodal??
# Shouldn't it resemble the prior??
# I think maybe the data plays a role in its unimodality.
```

Ex-3.7(a)

```
dta <- tibble(theta = seq(0, 1000)/1000) %>%
  mutate(posterior = dbeta(x = theta, shape1 = 3, shape2 = 14))

# graphics
dta %>%
  ggplot(aes(x = theta)) +
  geom_line(
    aes(y = posterior, color = "Posterior density"),
    linewidth = 1.5
  ) +
  geom_hline(
    aes(yintercept = 1, color = "Uniform prior"),
    linewidth = 1.2,
```

```r
    linetype = "dashed"
  ) +
  geom_vline(
    aes(xintercept = 2/15, linetype = "Posterior mode"),
    linewidth = 1
  ) +
  geom_vline(
    aes(xintercept = 3/17, linetype = "Posterior mean"),
    linewidth = 1
  ) +
  scale_color_manual(
    values = c(
      "Posterior density" = "red",
      "Uniform prior" = "blue"
    )
  ) +
  scale_linetype_manual(
    values = c(
      "Posterior mode" = "dashed",
      "Posterior mean" = "dotted"
    )
  ) +
  labs(
    x = expression(theta),
    y = "Density",
    color = "",
    linetype = "",
    title = "Posterior Density with Reference Lines"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "top"
  )
```



Ex-3.7(c)

```
y2 <- 0:278

# intermediate list 1
inter1 <- y2 * choose(n = 278, k = y2) * beta(a = (y2+3), b = (292-y2)) / beta(a = 3, b = 14)

# intermediate list 2
inter2 <- (y2^2) * choose(n = 278, k = y2) * beta(a = (y2+3), b = (292-y2)) / beta(a = 3, b = 14)

# 1st raw moment (Beta function is part of derive formula)
(
mean_Y2 <- sum(inter1)
)
```

```
## [1] 49.05882
```

```
# 2nd raw moment (Beta function is part of derive formula)
(
moment2 <- sum(inter2)
)
```

```
## [1] 3068.902
```

```r
# variance
(
var_Y2 <- moment2 - (mean_Y2^2)
)
```

```
## [1] 662.1338
```

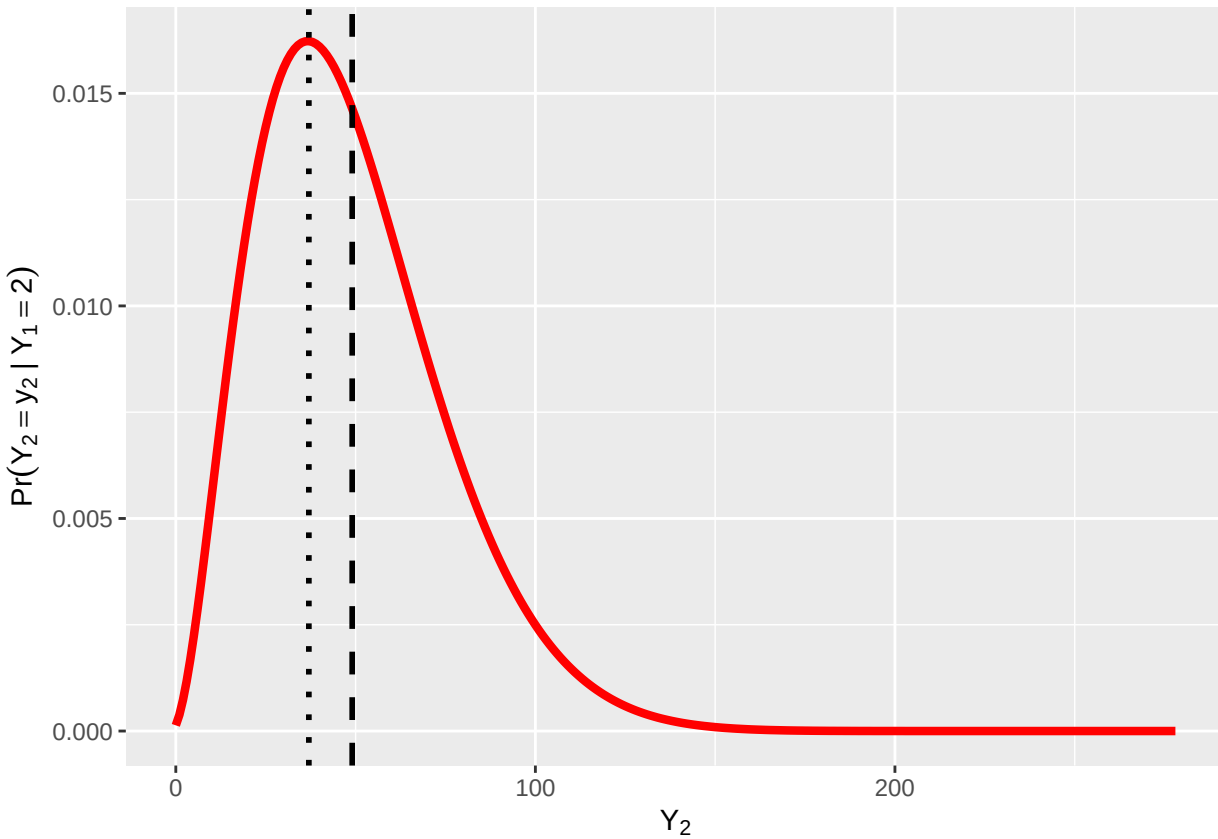
```r
# standard deviation
(
sd_Y2 <- var_Y2 %>% sqrt()
)
```

```
## [1] 25.73196
```

```r
# dataframe
dta <- tibble(Y2 = y2) %>%
  mutate(
    cond_prob = choose(n = 278, k = Y2)*beta(a = (Y2+3), b = (292-Y2))/beta(a = 3, b = 14)
  )

# mode
(
mode_Y2 <- dta %>%
  slice_max(cond_prob, n = 1) %>%
  pull(Y2)
)
```

```
## [1] 37
```

```r
# graphics (labels via plotmath)
dta %>%
  ggplot(aes(x = Y2, y = cond_prob)) +
  geom_line(linewidth = 1.5,
            color = "red") +
  geom_vline(xintercept = mean_Y2, # mean
             linewidth = 1,
             linetype = "dashed") +
  geom_vline(xintercept = mode_Y2, # mode
             linewidth = 1,
             linetype = "dotted") +
  xlab(expression(Y[2])) +
  ylab(expression(Pr(Y[2] == y[2] ~ "|" ~ Y[1] == 2)))
```



Ex-3.7(d)

Since the modes of $(Y_2 \mid \theta = \hat{\theta})$ and $(Y_2 \mid Y_1 = 2)$ are equal (although their means are far apart), the distributions are still **comparable** in terms of central tendency.

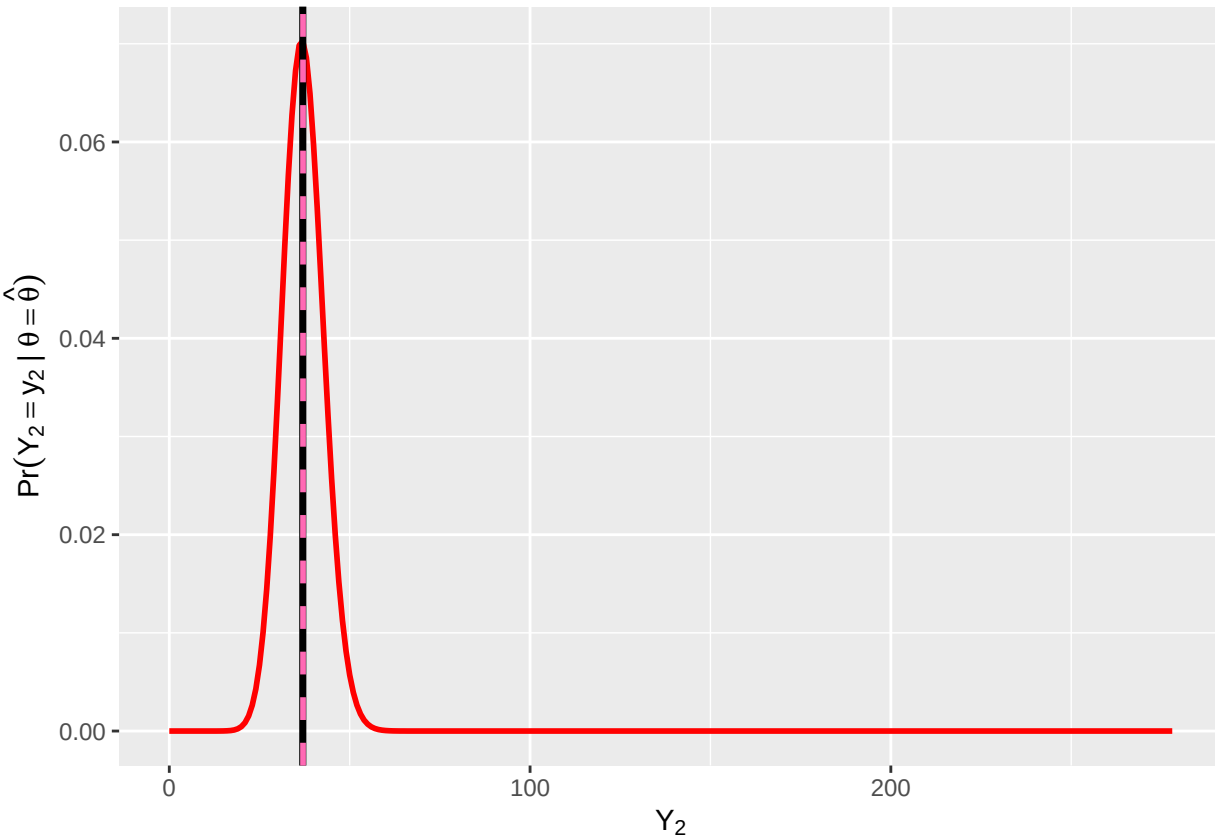
I would prefer to make predictions based of $(Y_2 \mid \theta = \hat{\theta})$ rather than $(Y_2 \mid Y_1 = 2)$ because the former has higher precision (lower variance) and it also incorporates data from the pilot study via $\hat{\theta}$ which is the **mode** and the **MLE** from $(\theta \mid Y_1 = 2)$.

```
y2 <- 0:278

# MLE and Posterior Mode
theta_hat <- 2/15

# sample size
n2 <- 278

# dataframe
dta <- tibble(Y2 = y2) %>%
  mutate(
    cond_prob = choose(n = n2, k = Y2)*(theta_hat^Y2)*(1-theta_hat)^(n2-Y2)
  )

# mean of Y2 conditioned on MLE of theta
```

```r
mean_Y2 <- sum(dta$Y2*dta$cond_prob)

# 2nd raw moment of Y2 conditioned on MLE of theta
moment2 <- sum((dta$Y2^2)*dta$cond_prob)

# variance of Y2 conditioned on MLE of theta
var_Y2 <- moment2 - mean_Y2^2

# standard deviation of Y2 conditioned on MLE of theta
sd_Y2 <- sqrt(var_Y2)

# mode of Y2 conditioned on MLE of theta
(
mode_Y2 <- dta %>%
  slice_max(cond_prob, n = 1) %>%
  pull(Y2)
)
```

```
## [1] 37
```

```r
# graphics (labels via plotmath)
dta %>%
  ggplot(aes(x = Y2, y = cond_prob)) +
  geom_line(linewidth = 1,
            color = "red") +
  geom_vline(xintercept = mode_Y2, # mode
             linewidth = 1.25,
             linetype = "solid") +
  geom_vline(xintercept = mean_Y2, # mean
             linewidth = 1,
             linetype = "dashed",
             color = "hotpink") +
  xlab(expression(Y[2])) +
  ylab(expression(Pr(Y[2] == y[2] ~ "|" ~ theta == hat(theta))))
```



Ex-3.9(a)

I'll do it in my spare time. It doesn't seem all that important to me although the Galenshore distribution is new to use, we've never heard of it before.